Convex Optimization Notes

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Contents

1 Jan 13: Introduction to Optimization **2**
1.1 Basic Definitions . . . 2

2 Jan 15: Convex Sets and Functions **3**
2.1 Convexity . . . 3
2.2 Local vs. Global Optima . . . 3

3 Jan 20: Convex Combinations, Hulls, and Cones **4**
3.1 Types of Combinations . . . 4
3.2 Norms and Convex Sets . . . 4
3.3 Optimality and the Normal Cone . . . 5

4 Jan 22: Optimality Conditions and Characterizations of Convexity **6**
4.1 Optimality and Operations on Convex Sets . . . 6
4.2 First and Second Order Conditions . . . 6
4.3 Subdifferentials . . . 7

5 Jan 27: Matrix Theory, Lipschitz Continuity, and Smoothness **8**
5.1 Gradient Monotonicity and SVD . . . 8
5.2 Matrix Norms . . . 8
5.3 Lipschitz Continuity and Smoothness . . . 8

6 Feb 3: Gradient Descent and Convergence **10**
6.1 Descent Directions and Step Sizes . . . 10
6.2 Convergence Rates . . . 10
6.3 Strong Convexity . . . 11

7 Feb 5: Subgradients and the Subgradient Method **12**
7.1 Review and Subgradient Calculus . . . 12
7.2 Subgradient Method . . . 12

8 Feb 10: Subgradient Method Analysis and Projections **13**
8.1 Subgradient Method Rates . . . 13
8.2 Polyak Stepsize and Feasibility . . . 14

9 Feb 12: Projected Subgradient Method **15**
9.1 Projections and Contractions . . . 15
9.2 Convergence and Examples . . . 15

10 Feb 14: Optimality Conditions and Proximal Gradient Descent **17**

11 Feb 19: Stochastic Gradient Descent **20**

12 Feb 24: **21**

13 March 10: Duality **23**
13.1 Primal and Dual LP: An Example . . . 23

1 Jan 13: Introduction to Optimization

1.1 Basic Definitions

Definition 1.1. *An **optimization problem** is defined as*

$$\min_{x \in \mathcal{C}} f_0(x) \quad \textit{where} \quad \forall i \in [m],\ f_i(x) \leq b_i$$

where $\mathcal{C}$ is the constraint set and $f_0 \colon \mathcal{C} \to \mathbb{R}$.

Definition 1.2. *The **standard form** of an optimization problem is*

$$\min_{x \in \mathcal{D}} f_0(x) \quad \textit{subject to} \quad \forall i \in [m],\ f_i(x) \leq 0; \quad \forall j \in [p],\ h_j(x) = 0$$

where $x \in \mathrm{dom}(f_i) \cap \mathrm{dom}(h_j) =: \mathcal{D}$.
f_0 is termed the objective function*, and x is the optimization variable.*
f_i and h_j are constraint functions.

Definition 1.3. *A **feasible solution** is any $x \in \mathcal{D}$ that satisfies all the constraints.*

Definition 1.4. *The **optimal solution** is $x^* \in \arg\min_{x \in \mathcal{D}} f_0(x)$.*

Remark 1.5. *If no feasible solution exists, we define $\min_{x \in \mathcal{C}} f(x) = +\infty$ and, similarly, $\max_{x \in \mathcal{C}} f(x) = -\infty$.*

2 Jan 15: Convex Sets and Functions

2.1 Convexity

Definition 2.1. *A set C is **convex** if $\forall x_1, x_2 \in C$ and $\forall \theta \in [0,1]$, we have $\theta x_1 + (1-\theta)x_2 \in C$.*

Definition 2.2. *A function $f\colon \mathcal{X} \to \mathbb{R}$ is **convex** if for all $x_1, x_2 \in \mathcal{X}$ and $\theta \in [0,1]$,*

$$f(\theta x_1 + (1-\theta)x_2) \leq \theta f(x_1) + (1-\theta) f(x_2) \tag{*}$$

Remark 2.3. *Convex functions are differentiable almost everywhere.*

Theorem 2.4. *If f is both convex and concave, then f is affine (i.e., linear plus constant).*

Definition 2.5. *f is **strictly convex** if $(*)$ holds with strict inequality $<$ whenever $x_1 \neq x_2$ and $\theta \in (0,1)$.*

Definition 2.6. *A **convex optimization problem** is one where f_0, f_i, and $\mathcal{C}$ are convex, and h_j are affine/linear. That is,*

$$\min_{x \in \mathcal{C}} f_0(x) \quad \text{such that} \quad f_i(x) \leq 0 \ \ \forall i \in [m], \quad h_j(x) = 0 \ \ \forall j \in [p]$$

Theorem 2.7. *The set of feasible solutions in Definition 2.6 is convex.*

2.2 Local vs. Global Optima

Theorem 2.8. *x^* is a **local minimizer** if there exists $\epsilon > 0$ such that for all y with $\|x^* - y\| \leq \epsilon$,*

$$f(x^*) \leq f(y)$$

Theorem 2.9. *x^* is a **local maximizer** if there exists $\epsilon > 0$ such that for all y with $\|x^* - y\| \leq \epsilon$,*

$$f(x^*) \geq f(y)$$

Theorem 2.10. *For any convex optimization problem, every local minimum is a global minimum.*

Proof. Suppose $\hat{x}$ is a local minimizer not equal to global minimizer x^*. Take ϵ as any witness to $\hat{x}$ being a local minimum. Let

$$y = \frac{\epsilon}{\|\hat{x} - x^*\|} x^* + \left(1 - \frac{\epsilon}{\|\hat{x} - x^*\|}\right) \hat{x}$$

Note: $\|\hat{x} - x^*\| \leq \epsilon$, otherwise $\hat{x}$ is not a local minimizer in that neighborhood.

$$y - \hat{x} = \frac{\epsilon(x^* - \hat{x})}{\|x^* - \hat{x}\|} \quad ; \quad \|y - \hat{x}\| = \epsilon$$

Since f is convex,

$$f(y) \leq \frac{\epsilon}{\|\hat{x} - x^*\|} f(x^*) + \left(1 - \frac{\epsilon}{\|\hat{x} - x^*\|}\right) f(\hat{x}) < f(\hat{x})$$

Thus, a contradiction. $\square$

3 Jan 20: Convex Combinations, Hulls, and Cones

3.1 Types of Combinations

Definition 3.1. *A **convex combination** of x_1, x_2 is $\theta_1 x_1 + \theta_2 x_2$ where $\theta_1, \theta_2 \geq 0$ and $\theta_1 + \theta_2 = 1$.*

Definition 3.2. *An **affine combination** of x_1, x_2 is $\theta_1 x_1 + \theta_2 x_2$ where $\theta_1 + \theta_2 = 1$.*

Definition 3.3. *A **linear combination** of x_1, x_2 is $\theta_1 x_1 + \theta_2 x_2$ where $\theta_1, \theta_2 \in \mathbb{R}$.*

Definition 3.4. *A **conic combination** of x_1, x_2 is $\theta_1 x_1 + \theta_2 x_2$ where $\theta_1, \theta_2 \geq 0$.*

Definition 3.5. *Given a set C, the **convex hull** of C is*

$$\operatorname{conv}(C) = \left\{ \sum_{i=1}^{k} \theta_i x_i \;\middle|\; x_i \in C,\ \theta_i \in [0,1],\ \sum_{i=1}^{k} \theta_i = 1 \right\}$$

Remark 3.6. *The following are true:*

1. $C \subseteq \operatorname{conv}(C)$
2. $\operatorname{conv}(C)$ *is convex*
3. *It is the smallest convex set containing C*
4. *If a convex set $S \supseteq C$ then $S \supseteq \operatorname{conv}(C)$*

Theorem 3.7. *Any closed convex set can be written as $\overline{\operatorname{conv}}(C)$ for some set C.*

Definition 3.8. *The **relative interior** of C is defined as*

$$\operatorname{relint}(C) = \{x \in C : \exists \epsilon > 0,\ B(x, \epsilon) \cap \operatorname{Aff}(C) \subseteq C\}$$

Definition 3.9. *C is a **cone** if $\alpha x \in C$ whenever $x \in C$ and $\alpha \geq 0$.*

Definition 3.10. *Given a set C, the **conic hull** of C is*

$$\operatorname{conic}(C) = \left\{ \sum_{i=1}^{k} \theta_i x_i : x_i \in C,\ \theta_i \geq 0 \right\}$$

Theorem 3.11. *The conic hull of C is the smallest convex cone containing C.*

3.2 Norms and Convex Sets

Definition 3.12. *The ℓ_p **norm** is defined as*

$$\|x\|_p = \left(\sum_{i=1}^{d} |x_i|^p \right)^{1/p}$$

Remark 3.13. *The following are true:*

1. *For $p \in (0,1)$: $\|x\|_p$ is not a convex function*
2. *For $p \geq 1$: $\|x\|_p$ is convex*
3. *For $p > 1$: $\|x\|_p$ is strictly convex*

Example 3.14. *Examples of convex sets:*

1. *Hyperplane: $\{x : a^T x = b\}$*
2. *Halfspace: $\{x : a^T x \leq b\}$*
3. *Polyhedron: $\{x \in \mathbb{R}^d : Ax \leq b,\ Cx = d\}$*
4. *Polytope: a bounded polyhedron.*

3.3 Optimality and the Normal Cone

Theorem 3.15. *A set S is **strictly convex** if for all $x_1 \neq x_2$ and $\theta \in (0,1)$, $\theta x_1 + (1-\theta)x_2 \in \operatorname{int}(S)$.*

Definition 3.16. *The **normal cone** is defined to be*

$$N_C(x) = \{g : g^T(y - x) \leq 0,\ \forall y \in C\}$$

Remark 3.17. *If $x \in \operatorname{int}(C)$ then $N_C(x) = \{0\}$.*

Theorem 3.18. *If f is differentiable, then x^* is optimal if and only if $-\nabla f(x^*) \in N_C(x^*)$.*

4 Jan 22: Optimality Conditions and Characterizations of Convexity

4.1 Optimality and Operations on Convex Sets

Theorem 4.1. *If f is convex and differentiable, and C is a convex set, then any optimal solution x^* to $\min_{x \in C} f(x)$ must satisfy $-\nabla f(x^*) \in N_C(x^*)$.*

Theorem 4.2. *The set of optimal solutions to a convex optimization problem is a convex set.*

Definition 4.3. *If C is convex then*

1. ***Translation:*** $C + a = \{x : x - a \in C\}$
2. ***Scaling:*** $\alpha C = \{x : \frac{x}{\alpha} \in C\}$
3. ***Intersection:*** *If $\{C_\alpha\}_{\alpha \in A}$ is a collection of convex sets, then $\bigcap_{\alpha \in A} C_\alpha$ is convex.*

Theorem 4.4. *The following are true:*

1. *If $C \subseteq \mathbb{R}^n$ is convex, $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, then $AC + b = \{Ax + b : x \in C\}$ is a convex set.*
2. *If $f(x) = Ax + b$, then $f^{-1}(C)$ is convex.*
3. *$C_1 + C_2 = \{x + y : x \in C_1,\, y \in C_2\}$ is convex.*
4. *If $C_1 \subseteq \mathbb{R}^m$, $C_2 \subseteq \mathbb{R}^n$ then $C_1 \times C_2 = \{(x, y) \in \mathbb{R}^{m+n} : x \in C_1,\, y \in C_2\}$ is convex.*
5. *For any $C \subseteq \mathbb{R}^n \times \mathbb{R}^m_{>0}$, define*

$$P(C) = \left\{ \left(\frac{x_1}{t}, \ldots, \frac{x_n}{t} \right) : (x_1, \ldots, x_n, t) \in C \right\}$$

If C is convex, so are $P(C)$ and $P^{-1}(C)$.

4.2 First and Second Order Conditions

Definition 4.5.

$$\operatorname{epi}(f) = \{(x, t) : x \in \operatorname{dom}(f),\, t \geq f(x)\}$$

Definition 4.6. ***First Order Definition of Convexity:*** *If f is differentiable, then f is convex if and only if for all $x, y \in \mathrm{dom}(f)$,*

$$f(y) \geq f(x) + \nabla f(x)^T (y - x)$$

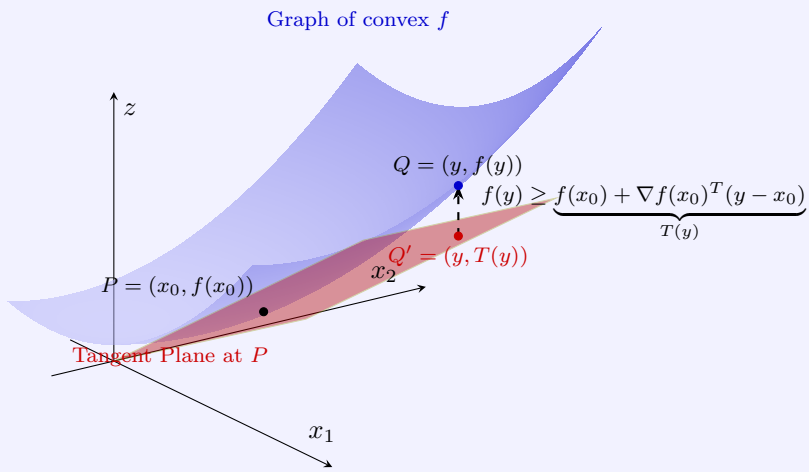

Figure 1: First Order Definition of Convexity

Second Order Definition of Convexity: *If f is twice differentiable, then f is convex if and only if $\nabla^2 f(x) \succeq 0$.*

Definition 4.7. *We say $A \succeq B$ if $A - B$ is positive semi-definite. This is equivalent to*

$$a^T \nabla^2 f(x) a \geq 0 \quad \text{for all } a \in \mathbb{R}^d$$

4.3 Subdifferentials

Definition 4.8. *The* ***subdifferential*** *is*

$$\partial f(x) = \left\{ g : f(y) \geq f(x) + g^T (y - x),\ \forall y \in C \right\}$$

Any such $g \in \partial f(x)$ is called a ***subgradient****.*
If f is differentiable at x then $\partial f(x) = \{\nabla f(x)\}$.

Theorem 4.9. *f is convex if and only if ∂f is non-empty.*

5 Jan 27: Matrix Theory, Lipschitz Continuity, and Smoothness

5.1 Gradient Monotonicity and SVD

Definition 5.1. *Gradient Monotonicity: If f is differentiable, then f is convex if and only if $\nabla f(x)$ is monotone. So we can conclude*

$$(\nabla f(x) - \nabla f(y))^T (x - y) \geq 0 \iff f \text{ is convex.}$$

Theorem 5.2. *For a $A \in \mathbb{R}^{m \times n}$ we can decompose*

$$A = U \Sigma V^t.$$

where $U \in \mathbb{R}^{m \times k}$ with orthonormal column, $\sigma \in \mathbb{R}^{k \times k}$ is a diagonal matrix with non-negative entries, and $V \in \mathbb{R}^{n \times k}$ with orthonormal column.

Definition 5.3. *(u_i, v_i, σ_i) form a s.v. triplet if $Av_i = \sigma_i u_i$ and $A^T u_i = \sigma_i v_i$*
$||u_i|| = ||v_i|| = 1$ and $\sigma_i \geq 0$

Theorem 5.4. *For any $A \in \mathbb{R}^{n \times n}$ then $A^T A$ is always positive semi definite*

Proof. We can write $A = V \sum U$ then $A^T A = V \sum U^T U \sum V^T = V \sum^2 V^T$ □

Definition 5.5. *Spectral Radius:* $\max_i \{|\lambda_i| : \lambda_i \in \Lambda(A)\} = \rho(A)$

Definition 5.6. *The norm must satisfy the following properties:*

1. $||A|| \geq 0$
2. $||\alpha A|| = |\alpha| ||A||$
3. $||A|| = 0$ *if and only if* $A = 0$
4. $||A + A'|| \leq ||A|| + ||A'||$

5.2 Matrix Norms

Definition 5.7. *Operator/Spectral Norm:*

$$||A||_{op} = ||A||_2 = \max_{||x||=1} ||Ax||_2.$$

Definition 5.8. *Frobenius Norm:*

$$||A||_F = \sqrt{\sum_{i,j} A_{ij}^2}.$$

5.3 Lipschitz Continuity and Smoothness

Definition 5.9. *f is L-Lipschitz if*

$$|f(x) - f(y)| \leq L||x - y||_2 \quad \forall x, y \in \text{dom}(f).$$

If f is differentiable then f is L-Lipschitz if and only if $|\nabla f| \leq L$

Definition 5.10. *Differentiable f is β-smooth if*

$$||\nabla f(x) - \nabla f(y)||_2 \leq \beta ||x - y||_2 \quad \forall x, y \in \mathrm{dom}(f).$$

If f is twice differentiable then f is β-smooth if and only if $\nabla^2 f(x) \preceq \beta I$.

Theorem 5.11. *If f is twice differentiable and β-smooth then*

$$f(y) \leq f(x) + \nabla f(x)^T (y - x) + \frac{\beta}{2} ||y - x||_2^2 \quad \forall x, y \in \mathrm{dom}(f).$$

6 Feb 3: Gradient Descent and Convergence

6.1 Descent Directions and Step Sizes

Definition 6.1. *"Descent Direction" any h for which $f(x+\eta h) \leq f(x)$.*

Theorem 6.2. *Assume f differentiable and β-smooth with $x_{t+1} \leftarrow x_t - \eta \nabla f(x_t)$.*

1. *If $\eta \leq \frac{2}{\beta}$ then $f(x_{t+1}) \leq f(x_t)$*
2. *If $\eta \leq \frac{1}{\beta}$ then $f(x_{t+1}) \leq f(x_t) - \frac{\eta}{2}||\nabla f(x_t)||_2^2$*

Proof. Recall

$$f(y) \leq f(x) + \nabla f(x)^T(y-x) + \frac{\beta}{2}||y-x||_2^2$$

and

$$||\nabla f(y) - \nabla f(x)||_2 \leq \beta ||y-x||_2$$

Let $y \leftarrow x+h$ then $f(x+h) \leq f(x) + \nabla f(x)^T h + \frac{\beta}{2}||h||_2^2$

1. If $h = -\frac{2}{\beta}\nabla f(x)$ then $\nabla f(x)^T h = -\frac{2}{\beta}||\nabla f(x)||_2^2$ and $\frac{\beta}{2}||h||_2^2 = \frac{2}{\beta}||\nabla f(x)||_2^2$ so $f(x+h) \leq f(x)$
2. If $h = -\frac{1}{\beta}\nabla f(x)$ then $\nabla f(x)^T h = -\frac{1}{\beta}||\nabla f(x)||_2^2$ and $\frac{\beta}{2}||h||_2^2 = \frac{1}{2\beta}||\nabla f(x)||_2^2$.

□

Theorem 6.3. *If f is differentiable, β smooth, and x^* be any optimizer. For any $k > 0$*

$$\min_{i=0,\dots,k} ||\nabla(f(x_i))||_2 \leq \sqrt{\frac{2\beta}{k}\left(\underbrace{f(x_0) - f(x^*)}_{\Delta_0}\right)}$$

Proof. AFSOC $\forall i \in \{0,\dots,k\}$, $||\nabla f(x_i)|| > \sqrt{\frac{2\beta}{k}\Delta_0}$.
Then we have

$$f(x_{i+1}) < f(x_i) - \frac{1}{2\beta}\left(\frac{2\beta\Delta_0}{k}\right) \qquad (\forall i \in \{0,\dots,k\})$$

Continuing down the chain we have

$$f(x_{k+1}) < f(x_0) - \Delta_0 < f(x^*)$$

Thus a contradiction. □

6.2 Convergence Rates

Theorem 6.4. *If f is β-smooth and convex, $\eta = \frac{1}{\beta}$ and x^* any minimum then*

$$f(x_k) - f(x^*) \leq \frac{\beta||x_0 - x^*||_2^2}{2k}$$

Proof. Observe that

$$||x_{t+1} - x^*||^2 = ||x_t - x^* - \eta\nabla f(x_t)||^2 = ||x_t - x^*||^2 - \eta^2||\nabla f(x_t)||^2 - 2\eta\nabla f(x_t)^T(x_t - x^*)$$

Rearranging and by convexity we have

$$\begin{aligned} f(x_t) - f(x^*) \leq \nabla f(x_t)^T (x_t - x^*) &= \frac{1}{2\eta}\left[\underbrace{||x_t - x^*||^2}_{\delta_t} - \underbrace{||x_{t+1} - x^*||^2}_{\delta_{t+1}} + \frac{\eta}{2}||\nabla f(x_t)||^2\right] \\ &\leq \frac{1}{2\eta}\left[\delta_t - \delta_{t+1} + 2\eta(f(x_t) - f(x_{t+1}))\right] \\ &= \frac{1}{2\eta}(\delta_t - \delta_{t+1}) + f(x_t) - f(x_{t+1}) \end{aligned}$$

So we have

$$f(x_{t+1}) - f(x^*) \leq \frac{\beta}{2}(\delta_t - \delta_{t+1})$$

Adding across all terms

$$\sum_{i=0}^{k-1} f(x_{i+1}) - f(x^*) \leq \frac{\beta}{2}\left(||x_0 - x^*||^2\right)$$

So we have the relation

$$k(f(x_k) - f(x^*)) \leq \sum_{i=0}^{k-1} f(x_{i+1}) - f(x^*) \leq \frac{\beta}{2}||x_0 - x^*||^2$$

□

6.3 Strong Convexity

Definition 6.5. *f is α-strongly convex if*

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y) - \frac{\alpha}{2}t(1-t)||x - y||_2^2.$$

Additionally, if f is differentiable then

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) + \frac{\alpha}{2}||y - x||_2^2.$$

If f is twice differentiable then f is α-strongly convex if and only if $\nabla^2 f(x) \succeq \alpha I$.

Theorem 6.6. *Assume f is β-smooth, α-strongly convex, $\eta = \frac{1}{\beta}$ then*

$$||x_k - x^*||^2 \leq \left(1 - \frac{1}{\gamma}\right)^k ||x_0 - x^*||^2.$$

We define the conditional number as $\gamma = \frac{\beta}{\alpha}$

Proof.

$$\frac{\alpha}{2}||x_k - x^*||^2 \leq \frac{\beta}{2}\left(||x_t - x^*||^2 - ||x_{t+1} - x^*||^2\right) \leq \left(1 - \frac{\alpha}{\beta}\right)||x_t - x^*||^2 \leq (1 - \frac{1}{\gamma})||x_k - x^*||^2$$

□

7 Feb 5: Subgradients and the Subgradient Method

7.1 Review and Subgradient Calculus

Recall 7.1. *From previous class*

1. $f(x+\eta h) \approx f(x) + \eta h^T \nabla f(x)$ *and when we set* $h = -\nabla f(x)$ *we have*

$$f(x - \eta \nabla f(x)) \approx f(x) - \eta ||\nabla f(x)||_2^2.$$

Backtracking Line Search: *Pick* $\gamma_1 \in (0,1)$ *and* $\gamma_2 \in (0,1)$*, start with initial step size* η*. While* $f(x-\eta\nabla f(x)) > f(x) - \gamma_1 \eta ||\nabla f(x)||_2^2$ *(Armijo condition not satisfied), set* $\eta \leftarrow \gamma_2 \eta$ *and retry. Once the condition is satisfied, use step size* η *to update* $x \leftarrow x - \eta \nabla f(x)$*.*

Recall 7.2. g *is a subgradient of* f *at* x *if*

$$f(y) \geq f(x) + g^T(y-x) \quad \forall y \in \mathrm{dom}(f).$$

Lemma 7.3. *The following are true:*

1. $\partial f(x)$ *makes sense for non convex functions too, could be empty*
2. *If* f *is convex, then for* $x \in \mathrm{RelInt}(\mathrm{dom}(f))$ *we have that* $\partial f(x)$ *is non-empty*
3. $\partial f(x)$ *is a convex*
4. *If* f *is convex and differentiable at* x *then* $\partial f(x) = \{\nabla f(x)\}$
5. *If* $\partial f(x)$ *is non-empty everywhere,* f *is convex.*

Theorem 7.4. *The following are true:*

1. $\partial(af) = a\partial f(x)$
2. $\partial(f+g) = \partial f(x) + \partial g(x)$
3. *If* $g(x) = f(Ax+b)$ *then* $\partial g(x) = A^T \partial f(Ax+b)$

Example 7.5. *For* $f(x) = \max_{i=1,\dots,n} f_i(x)$ *we have* $\partial f(x) = \mathrm{Conv}\left(\bigcup_{i=1,\dots,n} \partial f_i(x)\right)$

Example 7.6. *For* $f(x) = |x|$ *we have* $\partial f(x) = \mathrm{sign}(x) = \max\{-x, x\}$ *so* $\partial f(0) = [-1, 1]$*.*

Example 7.7. *Let* C *be a convex set and* $I_C(x) = \begin{cases} 0 & x \in C \\ \infty & x \notin C \end{cases}$ *is convex.*
Additionally, for $x \in C$*,* $\partial I_C(x) = N_C(x) = \{g : g^T(y-x) \leq 0, \forall y \in C\}$

Proof. For $x \in C$, $I_C(y) \geq I_C(x) + g^T(y-x)$ for all $y \in C$ then we have

$$0 \geq 0 + g^T(y-x).$$

□

7.2 Subgradient Method

Definition 7.8. ***Subgradient Method:*** $x_{t+1} \leftarrow x_t - \eta g_x$ *for some* $g_x \in \partial f(x)$

1. *Subgradients are in general not descent directions*
2. *The min norm subgradient is a descent direction*

Example 7.9. $f(x) = ||x||_2 = \sqrt{\sum_{i=1}^n x_i^2}$ *is convex, then for* $x \neq 0$ *we have* $\partial f(x) = \frac{x}{||x||_2}$*.*

$$\partial f(0) = \{g : ||g||_2 \leq 1\}.$$

8 Feb 10: Subgradient Method Analysis and Projections

8.1 Subgradient Method Rates

Definition 8.1. *For the subgradient method for* $\min_{x\in\mathbb{R}^n} f(x)$ *and* f *convex where*

$$x_{t+1} = x_t - \eta_t g_t \text{ where } g_t \in \partial f(x_t).$$

We define the best iterate as

$$x_T^{(\text{best})} = \arg\min_{i=0,\dots,T} f(x_i).$$

Theorem 8.2. *Assume* f *is* G*-lipschitz and convex. Let* $||x_0 - x^*|| \le R$ *then pick*

$$\eta_t = \frac{R}{G\sqrt{T}} \text{ guarantees that } f\left(x_T^{(\text{best})}\right) - f(x^*) \le \frac{GR}{\sqrt{T}}.$$

Theorem 8.3. *A convex function* f *is* G*-lipschitz iff*

$$||g_x|| \le G \ \forall x \in \text{dom}(f) \text{ and } \forall g_x \in \partial f(x).$$

Theorem 8.4. *For nonconvex, differentiable* f*,* f *being* G*-lipschitz iff*

$$||\nabla f(x)|| \le G \ \forall x \in \text{dom}(f).$$

Theorem 8.5. *Assume* f *is convex and* G*-Lipschitz, and that an optimal solution* x^* *exists with* $||x_0 - x^*|| \le R$ *for some* $R > 0$*.*
Pick $\eta_t \to 0$ *such that* $\sum_{t=1}^{\infty} \eta_t = \infty$ *and* $\sum_{t=1}^{T} \eta_t^2 < \infty$ *then*

$$f\left(x_T^{(\text{best})}\right) \to f(x^*) \text{ as } T \to \infty.$$

Theorem 8.6. *For the subgradient method with step sizes* $\{\eta_t\}_{t=1}^T$ *on a convex,* G*-Lipschitz function* f*, we have*

$$f\left(x_T^{(\text{best})}\right) - f(x^*) \ \le \ \frac{\|x_0 - x^*\|^2 + G^2 \sum_{t=1}^T \eta_t^2}{2\sum_{t=1}^T \eta_t}.$$

Verification of Theorem 8.2. Set $\eta_t = \frac{R}{G\sqrt{T}}$ for all t. With $\|x_0 - x^*\| \le R$:

$$\sum_{t=1}^{T} \eta_t = T \cdot \frac{R}{G\sqrt{T}} = \frac{R\sqrt{T}}{G},$$
$$\sum_{t=1}^{T} \eta_t^2 = T \cdot \frac{R^2}{G^2 T} = \frac{R^2}{G^2}.$$

Substituting into the bound (using $\|x_0 - x^*\|^2 \le R^2$):

$$f\left(x_T^{(\text{best})}\right) - f(x^*) \le \frac{R^2 + G^2 \cdot \frac{R^2}{G^2}}{2 \cdot \frac{R\sqrt{T}}{G}} = \frac{2R^2}{2R\sqrt{T}/G} = \frac{2R^2 \cdot G}{2R\sqrt{T}} = \frac{GR}{\sqrt{T}}.$$

Hence Theorem 8.2 holds with the $O(1/\sqrt{T})$ rate. □

Proof.

$$\begin{aligned} ||x_{t+1} - x^*||^2 &= ||x_t - \eta_t g_t - x^*||^2 \\ &= ||x_t - x^*||^2 - 2\eta_t g_t^T (x_t - x^*) + \eta_t^2 ||g_t||^2 \\ &\le ||x_t - x^*||^2 + 2\eta_t (f(x^*) - f(x_t)) + \eta_t^2 G^2 \end{aligned} \tag{*}$$

Last step follows from $f(x^*) \geq f(x_t) + g_t^T(x^* - x_t) \iff -g_t^T(x_t - x^*) \geq f(x^*) - f(x_t)$.
Adding $(*)$ up from $0, ..., T-1$ we have

$$||x_T - x^*||^2 \leq ||x_0 - x^*||^2 + G^2 \sum_{t=0}^{T-1} \eta_t^2 + 2 \sum_{t=0}^{T-1} \eta_t \left(f(x^*) - f(x_t)\right).$$

So

$$2 \sum_{t=0}^{T-1} \eta_t \left(f(x_T^{(\text{best})}) - f(x^*)\right) \leq 2 \sum_{t=0}^{T-1} \eta_t \left(f(x_t) - f(x^*)\right) \leq ||x_0 - x^*||^2 + G^2 \sum_{t=0}^{T-1} \eta_t^2.$$

□

8.2 Polyak Stepsize and Feasibility

Theorem 8.7. *Polyak's Stepsize: If $f(x^*)$ is known,*

$$\eta_t = \frac{f(x_t) - f(x^*)}{||g_t||^2}.$$

Theorem 8.8. *Given $C_1, ..., C_k$ convex sets find $x^* \in \bigcap_{i=1}^k C_i$.*

Proof. Define $f_i(x) = \min_{y \in C_i} ||x - y||^2 = \text{dist}(x, C_i)$ and $f(x) = \max_{i=1,...,k} f_i(x)$.
If $x^* \in C_1 \cap ... \cap C_k$ then $f(x^*) = 0$.
Recall: $\partial f(x) = \text{Conv}\left(\bigcup_{i=1,...,k} \partial f_i(x)\right)$.
Let $P_C(x) = \arg\min_{y \in C} ||x - y||$

Lemma 8.9. *u is the projection of x onto C iff $\langle x - u, y - u \rangle \leq 0 \ \forall y \in C$*

We have $f_i(x) = ||x - P_{C_i}(x)||^2$ so $\partial f_i(x) = \frac{x - P_{C_i}(x)}{|x - P_{C_i}(x)||}$ if $x \neq 0$.

$$x_{t+1} = x_t - f(x_t) \cdot \frac{x_t - P_{C_i}(x_t)}{||x_t - P_{C_i}(x_t)||} = P_{C_i}(x_t).$$

□

9 Feb 12: Projected Subgradient Method

9.1 Projections and Contractions

Definition 9.1. *For* $\min_{x\in C} f(x)$ *we define the projected subgradient method to be*

$$\begin{cases} y_{t+1} = x_t - \eta_t g_t \\ x_{t+1} = P_C(y_t) \end{cases}.$$

Theorem 9.2. $P_C(x) = \arg\min_{y\in C} ||x-y||^2$

Proof. We have $z = P_c(x)$ iff $\forall y \in C$, $\langle x-z, y-z\rangle \leq 0$. Then for

$$\underbrace{-\nabla f(z)}_{x-z} \in \underbrace{N_c(z)}_{\{g:g^T(y-z)\leq 0\ \forall y\in C\}}.$$

so we have $(x-z)^T(y-z) \leq 0$ □

Lemma 9.3. *Projections are contractions:*

$$||\underbrace{P_C(x_1)}_{z_1} - \underbrace{P_C(x_2)}_{z_2}||_2 \leq ||x_1 - x_2||_2.$$

Proof. We have

1. $\forall z_2 \in C, \langle x_1 - z_1, z_2 - z_1\rangle \leq 0$
2. $\forall z_1 \in C, \langle x_2 - z_2, z_1 - z_2\rangle \leq 0$
3. Adding these two inequalities we have

$$\langle x_1 - z_1, z_2 - z_1\rangle + \langle x_2 - z_2, z_1 - z_2\rangle = \langle x_1 - x_2 + z_2 - z_1, z_2 - z_1\rangle \leq 0.$$

So

$$\langle x_1 - x_2, z_2 - z_1\rangle + ||z_2 - z_1||_2^2 \leq 0 \iff ||z_2 - z_1||_2^2 \leq \langle x_1 - x_2, z_2 - z_1\rangle \leq ||x_1 - x_2||||z_2 - z_1||.$$

So we have our result

$$\|z_1 - z_2\| \leq ||x_1 - x_2||.$$

□

9.2 Convergence and Examples

Theorem 9.4. *Rates for PGD are identical to GD and similarly PSGM is identical to SGM.*

Proof.

$$||x_{t+1} - x^*||^2 = ||P_C(x_t - \eta_t g_t) - x^*||^2 = ||P_C(x_t - \eta_t g_t) - P_C(x^*)|| \leq ||x_t - \eta_t g_t - x^*||^2.$$

□

Example 9.5. *Consider*

$$\min_{x\in\mathbb{R}^d} \frac{1}{2}||Ax - b||_2^2 \text{ where } n \geq d \text{ and } A \text{ full rank}.$$

When we solve this by gradient descent then the rate would be

$$\left(1 - \frac{\lambda_{min}}{\lambda_{max}}\right)^k.$$

Proof. We have $\nabla f(x) = A^T(Ax - b)$ and $\nabla^2 f(x) = A^T A$. So f is quadratic with Hessian $A^T A$. The eigenvalues of $A^T A$ satisfy $\lambda_{\min} = \lambda_{\min}(A^T A)$ and $\lambda_{\max} = \lambda_{\max}(A^T A)$. Hence f is α-strongly convex with $\alpha = \lambda_{\min}$ and β-smooth with $\beta = \lambda_{\max}$. Taking $\gamma = \beta/\alpha = \lambda_{\max}/\lambda_{\min}$ and $\eta = 1/\beta$, the strongly convex convergence theorem gives

$$||x_k - x^*||^2 \leq \left(1 - \frac{1}{\gamma}\right)^k ||x_0 - x^*||^2 = \left(1 - \frac{\lambda_{\min}}{\lambda_{\max}}\right)^k ||x_0 - x^*||^2.$$

□

Example 9.6. *Consider a similar problem*

$$\min_{x \geq 0} \frac{1}{2}||Ax - b||_2^2 \text{ where } n \geq d \text{ and } A \text{ full rank .}$$

We can let $y_{t+1} \leftarrow x_t - \eta_t(Ax_t - b)$ *and* $x_{t+1} \leftarrow \max(0, y_{t+1})$.

Example 9.7. *Example of Fast Projection:*

$$C = \left\{Ay + b | y \in \mathbb{R}^\ell\right\} \subseteq \mathbb{R}^n \text{ and } P_C(x) = b + A(A^T A)^{-1} A^T (x - b).$$

Proof.

$$y^* = \arg\min_{y \in \mathbb{R}^\ell} \frac{1}{2}||x - (Ay + b)||_2^2 \implies A^T(Ay^* + b) - x = 0 \implies A^T A y^* = A^T(x - b) \implies y^* = (A^T A)^{-1} A^T (x - b).$$

So $P_C(x) = Ay^* + b$ □

Example 9.8.

$$C = \{y : Ay = b\} \subseteq R^d \text{ and } P_c(x) = x + A^T(AA^T)^{-1}(b - Ax).$$

10 Feb 14: Optimality Conditions and Proximal Gradient Descent

Theorem 10.1. *Consider* $\min_{x\in\mathbb{R}^d} f(x)$, *$f$ not assumed convex. Then x^* is optimal on* $\mathrm{dom}(f)$ *iff* $0 \in \partial f(x^*)$.

Proof. ($\Longleftarrow$) We have

$$f(y) \geq f(x^*) + \langle g_{x^*}, y - x^* \rangle . \qquad (\forall y \in \mathrm{dom}(f) \text{ and } g_x^* \in \partial f(x^*))$$

Since $0 \in \partial f(x^*)$ we have $f(y) \geq f(x^*)$ for all $y \in \mathrm{dom}(f)$. So x^* is optimal.
($\Longrightarrow$)

$$f(y) \geq f(x^*) + \langle 0, y - x^* \rangle \implies 0 \in \partial f(x^*).$$

□

Theorem 10.2. *f is convex and differentiable with C convex then*

$$x^* \in \arg\min_{x\in C} f(x) \iff -\nabla f(x^*) \in N_C(x^*).$$

Theorem 10.3. *If f is convex and C convex then*

$$x^* \text{ is optimal for } f \text{ in } C \iff 0 \in \partial f(x^*) + N_C(x^*).$$

Proof. We can rewrite the objective as

$$x^* \in \arg\min_{x\in C} f(x) = \arg\min_{x\in\mathbb{R}^d} f(x) + I_C(x) \iff 0 \in \partial f(x^*) + N_C(x^*).$$

□

Example 10.4. *Soft Thresholding: Consider the optimization problem with an L_2 data fidelity term and an L_1 regularization term (Lasso):*

$$x^* = \arg\min_{z\in\mathbb{R}^d} \left(\frac{1}{2}\|z - y\|_2^2 + \lambda\|z\|_1 \right).$$

Since the objective function is separable (a sum of terms for each component), we can solve for each coordinate x_j^* independently. The problem reduces to finding scalar x_j^* that minimizes:

$$f(z_j) = \frac{1}{2}(z_j - y_j)^2 + \lambda|z_j|.$$

The optimality condition for a convex function requires that 0 is in the subdifferential of the objective at the solution x_j^*:

$$0 \in \partial \left(\frac{1}{2}(x_j^* - y_j)^2 + \lambda|x_j^*| \right).$$

Taking the derivative of the quadratic term and the subdifferential of the absolute value, we get:

$$0 \in (x_j^* - y_j) + \lambda\partial|x_j^*|.$$

where the subdifferential $\partial|x|$ is defined as:

$$\partial|x| = \begin{cases} \{1\} & \text{if } x > 0 \\ \{-1\} & \text{if } x < 0 \\ [-1, 1] & \text{if } x = 0 \end{cases}$$

We analyze the three possible cases for the solution x_j^*:

1. ***Case 1*** *($x_j^* > 0$):* *The subgradient is $\{1\}$.*
$$x_j^* - y_j + \lambda = 0 \implies x_j^* = y_j - \lambda.$$
For this to be consistent with our assumption $x_j^* > 0$, we must have $y_j - \lambda > 0$, or $y_j > \lambda$.
2. ***Case 2*** *($x_j^* < 0$):* *The subgradient is $\{-1\}$.*
$$x_j^* - y_j - \lambda = 0 \implies x_j^* = y_j + \lambda.$$
For this to be consistent with $x_j^* < 0$, we must have $y_j + \lambda < 0$, or $y_j < -\lambda$.
3. ***Case 3*** *($x_j^* = 0$):* *The subgradient is the interval $[-1, 1]$. The condition becomes $0 \in -y_j + \lambda[-1, 1]$, which means $y_j \in [-\lambda, \lambda]$. This holds whenever $|y_j| \leq \lambda$.*

Combining these cases yields the closed-form ***Soft Thresholding Operator****, denoted as $\mathcal{S}_\lambda(y)$:*
$$x_j^* = \mathcal{S}_\lambda(y_j) = \begin{cases} y_j - \lambda & \text{if } y_j > \lambda \\ y_j + \lambda & \text{if } y_j < -\lambda \\ 0 & \text{if } |y_j| \leq \lambda \end{cases}$$

Or more compactly using the sign function:
$$x^* = sign(y) \max(|y| - \lambda, 0).$$

Theorem 10.5. *Proximal Gradient Descent: Our objective is*
$$\min_{x \in \mathbb{R}^d} \underbrace{g(x)}_{\substack{convex \\ diff}} + \underbrace{h(x)}_{\substack{convex \\ non-diff \\ simple}}.$$
Our update rules are
$$y^{t+1} \leftarrow x^t - \eta_t \nabla g(x^t)$$
$$x^{t+1} \leftarrow prox_{\eta_t, h}(y^{t+1}) := \arg\min_{z \in \mathbb{R}^d} \frac{1}{2\eta_t} ||y^t - z||_2^2 + h(z)$$
When $g(x) = ||Ax - b||_2^2$, $h(x) = \lambda ||x||_1$ then this is Proximal GD called ISTA.

Remark 10.6. *The intuition is that*
$$x_{t+1} \leftarrow \arg\min_z g(x_t) + \nabla g(x_t)^T (z - x_t) + \frac{1}{2\eta_t} ||z - z_t||_2^2 + h(z) = \arg\min_z \frac{1}{2\eta_t} ||z - (x_t - \eta_t \nabla g(x_t))||_2^2 + h(z).$$

Theorem 10.7. *For $p(x) = Prox_{\eta,h}(x) = \arg\min_z \frac{1}{2\eta} ||x - z||_2^2 + h(z)$ we have*

1. $||p(x) - p(y)||_2^2 \leq \langle x - y, p(x) - p(y) \rangle$
2. $||p(x) - p(y)|| \leq ||x - y||$

(1) $\implies$ (2) *by the Cauchy-Schwarz inequality.*

Proof. We have
$$\frac{1}{\eta}(x - p(x)) \in \partial h(p(x)) \text{ and } \frac{1}{\eta}(y - p(y)) \in \partial h(p(y)).$$
By subgradient monotonicity we have
$$\langle g_x - g_y, x - y \rangle \geq 0 \text{ for all } x, y \text{ and } g_x \in \partial f(x), g_y \in \partial f(y).$$
So
$$\left\langle \frac{1}{\eta}(y - p(y)) - \frac{1}{\eta}(x - p(x)), p(y) - p(x) \right\rangle \geq 0 \iff \langle y - x, p(y) - p(x) \rangle \geq ||p(y) - p(x)||_2^2.$$
□

Theorem 10.8. *Define* $G_\eta(x) = \frac{x - prox_{\eta,h}(x - \eta\nabla g(x))}{\eta}$ *then the following are true:*

1. $G_\eta(x) = 0 \iff x = prox_{\eta,h}(x - \eta\nabla g(x))$
2. $G_\eta(x^*) = 0 \iff 0 \in \nabla g(x^*) + \partial h(x^*) \iff x^*$ *is optimal* $\iff x^*$ *is a fixed point of prox*

Proof. By definition $\forall \tilde{x}$ then

$$\tilde{x} - \eta G_\eta(\tilde{x}) = prox_{\eta,h}(\underbrace{\tilde{x} - \eta\nabla g(\tilde{x})}_{x}).$$

$$\begin{aligned} u := prox_{\eta,h}(x) &\iff \frac{1}{\eta}(x - u) \in \partial h(u) \\ &\iff \tilde{x} - \eta\nabla g(\tilde{x}) - \tilde{x} - \eta G_\eta(\tilde{x}) \in \eta\partial h(\tilde{x} - \eta G_\eta(\tilde{x})) \\ &\iff G_\eta(\tilde{x}) \in \nabla g(\tilde{x}) + \partial h(\tilde{x} - \eta G_\eta(\tilde{x}))\ \forall \tilde{x} \end{aligned}$$

Thus we imply 1 by choosing $\tilde{x} = x^*$ □

11 Feb 19: Stochastic Gradient Descent

Definition 11.1. *Stochastic Gradient Descent: We have access to $g(x_t, \zeta_t)$ unbiased for $\nabla f(x_t)$ or an element of $\partial f(x_t)$.*

$$\mathbb{E}_\zeta[g(x,\zeta)] = \nabla f(x) \in \partial f(x).$$

Our algorithm performs update

$$x^{t+1} \leftarrow x^t - \eta_t g(x^t, \zeta_t).$$

Example 11.2. *$R_n(\theta) = \frac{1}{n}\sum_{i=1}^n \ell(f_\theta(x_i, y_i))$, if n is large, computing ∇R_n is expensive.*

$$g_i = \nabla \ell(f_\theta(x_i), y_i) \text{ is unbiased for } \nabla R_n(\theta) \text{ if } i \sim \text{Unif}\{1, ..., n\}.$$

Theorem 11.3. *Assume f is convex, $||x_0 - x^*|| = R$ and $\mathbb{E}[||g(x_t, \zeta_t)||_2^2] \leq G^2$ then choosing $\eta = \frac{R}{G\sqrt{k}}$ yields*

$$\mathbb{E}\left[f\left(\frac{1}{k}\sum_{i=1}^k x_i\right) - f(x^*)\right] \leq \frac{RG}{\sqrt{k}}.$$

Proof. Define $g_t = \mathbb{E}[\tilde{g}_t]$ then we have

$$||x_{t+1} - x^*||^2 = ||x_t - \eta_t \tilde{g}_t - x^*||^2 = ||x_t - x^*||^2 - 2\eta_t \tilde{g}_t^T(x_t - x^*) + \eta_t^2 g_t^2.$$

So

$$\begin{aligned}
\mathbb{E}[||x_{t+1} - x^*||_2^2 \mid x_t] &\leq ||x_t - x^*||^2 - 2\eta_t g_t^T(x_t - x^*) + \eta_t^2 G^2 \\
&\leq ||x_t - x^*||^2 - 2\eta_t(f(x_t) - f(x^*)) + \eta_t^2 G^2 \\
\mathbb{E}[||x_{t+1} - x^*||_2^2] &\leq \mathbb{E}[||x_t - x^*||^2] - 2\eta_t \mathbb{E}[f(x_t) - f(x^*)] + \eta_t^2 G^2
\end{aligned}$$

So we can conclude

$$0 \leq \mathbb{E}[||X_{t+1} - x^*||_2^2] \leq R^2 - 2\sum_{i=1}^k \mathbb{E}[f(x_i) - f(x^*)] + \sum_{t=1}^k \eta_t^2 G^2.$$

Rearranging we have

$$2\sum_{i=1}^k \eta_t \mathbb{E}[f(x_i) - f(x^*)] \leq R^2 + \sum_{t=1}^k \eta_t^2 G^2.$$

For $\eta_t = \eta$ we have

$$\frac{1}{k}\sum_{i=1}^k \mathbb{E}[f(x_i) - f(x^*)] \leq \frac{1}{k}\left(\frac{R^2}{2\eta} + \frac{\eta k G^2}{2}\right).$$

So $\eta^2 = \frac{R^2}{kG^2} \implies \eta = \frac{RG}{\sqrt{k}}$.
Proof concludes by $f\left(\frac{1}{k}\sum_{i=1}^k x_i\right) \leq \frac{1}{k}\sum_{i=1}^k f(x_i)$. □

Theorem 11.4. *Assume f is α-strongly convex. For any $\eta < \frac{1}{\alpha}$ we have*

$$\mathbb{E}\left(||x_k - x^*||_2^2\right) \leq (1 - \eta\alpha)^k ||x_0 - x^*||_2^2 + \frac{G\eta}{\alpha}.$$

12 Feb 24:

Recall 12.1. $\min_{x\in\mathbb{R}^d} g(x)+h(x)$ *where g is β-smooth and convex and $h(x)$ is non-smooth, convex, and simple.*

$$Prox_{\eta,h}(x) = \arg\min_{z\in\mathbb{R}^d} \frac{1}{2\eta}||x-z||_2^2 + h(z).$$

We define the update rules as

$$y_{t+1} \leftarrow x_t - \eta_t \nabla g(x_t) \text{ and } x_{t+1} \leftarrow Prox_{\eta_t,h}(y_{t+1}).$$

Lemma 12.2. *Descent Lemma For $\eta \leq \frac{1}{\beta}$ and for any z,*

$$f(x-\eta G_\eta(x)) \leq f(z) + G_\eta(x)^T(x-z) - \frac{\eta}{2}||G_\eta(x)||_2^2 - \frac{\alpha}{2}||x-z||_2^2.$$

Remark 12.3. *When $z = x$ we have*

$$f(x_{t+1}) \leq f(x_t) - \frac{\eta}{2}||G_\eta(x_t)||_2^2.$$

Theorem 12.4. *For $\eta = \frac{1}{\beta}$ we have*

$$f(x_k) - f(x^*) \leq \frac{\beta||x_0 - x^*||_2^2}{2k}.$$

Proof. From the Descent Lemma, if we let $z = x^*$ we have

$$f(x_{t+1}) \leq f(x^*) + G_\eta(x_t)^T(x_t - x^*) - \frac{\eta}{2}||G_\eta(x_t)||_2^2.$$

$$\begin{aligned}||x_{t+1} - x^*||_2^2 &= ||x_t - x^*|| - 2\eta G_\eta(x_t)^T(x_t - x^*) + \eta^2||G_\eta(x_t)||_2^2 \\ &\leq ||x_t - x^*||_2^2 + 2\eta(f(x^*) - f(x_{t+1}))\end{aligned}$$

Rearranging we have

$$f(x_k) - f(x^*) \leq f(x_{t+1}) - f(x^*) \leq \frac{1}{2\eta}\left(||x_t - x^*||_2^2 - ||x_{t+1} - x^*||_2^2\right).$$

Telescoping for $t = 0, ..., k-1$ we have

$$k(f(x_k) - f(x^*)) \leq \frac{1}{2\eta}\left(||x_0 - x^*||_2^2 - ||x_{k+1} - x^*||_2^2||_2^2\right).$$

So

$$f(x_k) - f(x^*) \leq \frac{\beta||x_0 - x^*||_2^2}{2k}.$$

□

Definition 12.5. *Let $f : C \to \mathbb{R}$ be non-convex then g is called the convex envelope of f if it is the largest convex function that lies below f. Formally $g(x) \leq f(x)$ for all $x \in C$.*

Example 12.6. *Matrix Completion (Netflix Problem): Given a $n \times m$ matrix M^* with $O(n+m)$ non-zero entries. Let $\mathcal{I}$ be the observed indices then the set of observed values is $\{y_{i,j}|(i,j) \in \mathcal{I}\}$. As of right now we can't recover the matrix so if we instead assume M^* is a low rank matrix then we can write*

$$\hat{M}^* = \arg\min_X \sum_{(i,j)\in\mathcal{I}} (M_{i,j} - y_{i,j})^2 + \lambda||M||_*.$$

Then

$$Prox_{\eta,*}(L) = \arg\min_M \frac{1}{2}||M - L||_2^2 + \eta||M||_* = U\Sigma_\eta V^T. \qquad (\text{where} L = U\Sigma V^T)$$

Theorem 12.7. *Let $s(x) = rank(X)$, $C = \{X : ||X||_2 \leq 1\}$ with $t(X) \leq s(X)$ for all $X \in C$ then*

$$t(X) = ||X||_* = \sum_{i=1}^{r} \sigma_i = tr(\sqrt{X^T X}) \leq rank(x).$$

13 March 10: Duality

Definition 13.1. *At a high level, it gives you, every dual feasible point gives a provable lower bound of the original primal problem, even if the original (primal) problem is non-convex. (Weak Duality)*

Definition 13.2. *A LP is simply an optimization problem*

$$\min_{x\in\mathbb{R}^d} c^T x \text{ subject to } Ax = b, Gx \leq h.$$

13.1 Primal and Dual LP: An Example

For any $a, b, c \geq 0$ with $a + b = p$ and $a + c = q$, we obtain the lower bound

$$\min_{x,y}\{px + qy : x + y \geq 2,\ x, y \geq 0\} \geq 2a$$

What's the best we can do? Maximize our lower bound over all feasible a, b, c:

Primal LP

$$\begin{aligned} \min_{x,y} \quad & px + qy \\ \text{subject to} \quad & x + y \geq 2 \\ & x, y \geq 0 \end{aligned}$$

Dual LP

$$\begin{aligned} \max_{a,b,c} \quad & 2a \\ \text{subject to} \quad & a + b = p \\ & a + c = q \\ & a, b, c \geq 0 \end{aligned}$$

Remark 13.3. *In this example, we have*

- *The number of dual variables equals the number of primal constraints (3 in this example).*
- *Weak duality:* $\min_{x,y} px + qy \geq \max_{a,b,c} 2a$.

Theorem 13.4. *Given* $c \in \mathbb{R}^n, A \in \mathbb{R}^{m\times n}, b \in \mathbb{R}^m, G \in \mathbb{R}^{r\times n}, h \in \mathbb{R}^r$ *then for the primal LP*

$$\min_{x\in\mathbb{R}^n} c^T x \text{ subject to } Ax = b, Gx \leq h.$$

Then the dual LP is

$$\max_{u\in\mathbb{R}^m, v\in\mathbb{R}^r} -b^T u - h^T v \text{ subject to } -A^T u - G^T v = c, v \geq 0.$$

Proof. For any $u \in \mathbb{R}^m, v \in \mathbb{R}^r$, $v \geq 0$ and x primal feasible, we have

$$\begin{aligned} u^T(Ax - b) + v^T(Gx - h) &\leq 0 \\ u^T Ax - u^T b + v^T Gx - v^T h &\leq 0 \\ (u^T A + v^T G)x &\leq u^T b + v^T h \\ (A^T u + G^T v)^T x &\leq b^T u + h^T v \\ (-A^T u - G^T v)^T x &\geq -b^T u - h^T v \end{aligned}$$

If $c = -A^T u - G^T v$ and x primal feasible

$$c^T x \geq -b^T u - h^T v.$$

and we have a lower bound on the primal feasible optimal value $c^T x^*$. □

Remark 13.5. *The dual is also a linear program. It is a maximization program*

Remark 13.6. *Each constarnt in the primal, yields a variable in the dual*